

improve service efficiency while reducing operational costs.

Within airports, AI is reshaping identity verification and boarding processes. Facial recognition systems and biometric check-in solutions are increasingly adopted at major international hubs such as Singapore Changi, Heathrow, and Hartsfield-Jackson Atlanta International Airport. These technologies use AI-driven computer vision to rapidly match passengers' facial data with passport and ticket records, reducing dependence on physical documents and significantly cutting waiting times at security screening and boarding gates.

In addition, recommendation engines that analyse passenger travel history and preferences can suggest relevant add-ons, such as lounge access, priority boarding, or meal options, during booking and check-in stages. This level of personalisation enhances passenger engagement while improving overall service efficiency.

Weather prediction and impact analysis

Weather remains one of the most unpredictable and influential factors in aviation operations. Phenomena such as turbulence, wind shear, thunderstorms, and fog can cause delays, route deviations, or flight cancellations. Artificial intelligence is reshaping aviation meteorology by enabling airlines and airports to anticipate and respond to changing weather conditions with far greater accuracy.

AI systems can analyse vast meteorological datasets—including satellite imagery, radar data,



Facial recognition at airports (Courtesy: aviationtoday.com)

wind profiles, humidity levels, and atmospheric pressure—much faster than traditional forecasting tools. By applying machine learning techniques to identify complex patterns, AI improves both short-term and long-range weather forecasts, providing airlines with additional time to prepare operational responses.

A notable example is Lufthansa's collaboration at Zurich Airport, where AI-driven systems were deployed to improve wind pattern forecasting accuracy. By feeding real-time data into predictive models, Lufthansa achieved a reported 40% improvement in wind forecast precision, a critical factor during take-off and landing operations. This improvement supported better runway utilisation, reduced fuel burn during holding patterns, and minimised last-minute delays caused by unexpected wind shifts.

AI-based turbulence prediction tools are also being adopted globally. Organisations such as The

Weather Company, a subsidiary of IBM, are developing AI-enhanced aviation weather platforms that deliver real-time turbulence alerts, enabling pilots to proactively adjust routes or altitudes. These capabilities enhance passenger comfort and reduce the risk of injuries caused by sudden in-flight turbulence.

In addition, AI-powered fog forecasting systems support ground operations by enabling more effective scheduling in low-visibility conditions. Several airports across Asia are trialling AI models capable of predicting fog formation up to six hours in advance—well beyond the capabilities of conventional meteorological methods.

Artificial intelligence is reshaping the aviation sector by delivering advanced solutions that enhance safety, optimise operational efficiency, and improve the passenger experience. From air traffic management and autonomous aircraft to personalised services and weather forecasting, AI is embedded across multiple layers of aviation operations. As these technologies continue to mature, their role will expand further, enabling more informed decision-making, reducing environmental impact, and strengthening the long-term efficiency and resilience of the aviation industry. **EFY**



CityAirbus (Courtesy: euronews.com)